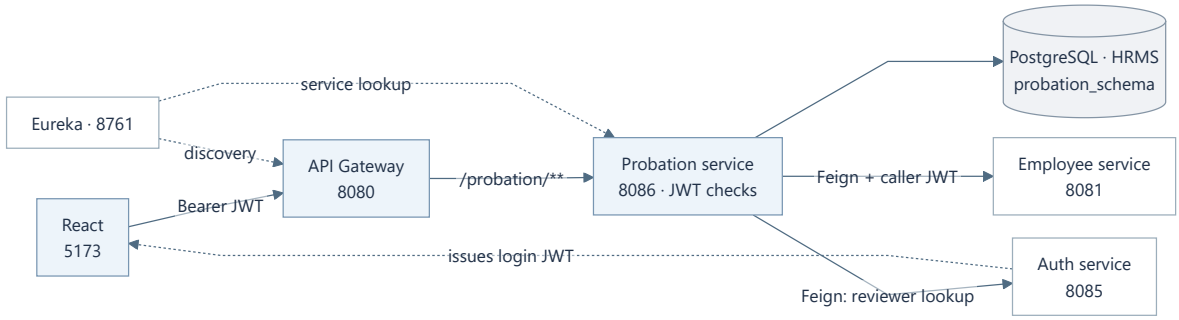
Probation Service Explained

Probation assessment reviews an employee's work during an initial employment period. It helps companies record evidence, identify development needs and make accountable decisions. HRMS **probation-service** coordinates this review, protects confidential feedback and publishes the authorized outcome. It does not change employment status, payroll or accounts.

1. Where the service fits



JWT is a signed login token. Gateway strips `/api`; probation verifies JWTs. Dashed arrows: discovery or token issuance.

2. How a request becomes a saved assessment

Main classes	What they do
`ProbationController`	Validates request DTOs and delegates HTTP actions.
`ProbationServiceImpl` `WorkflowRules`	Checks caller, state and version; saves the change and audit together in one transaction.
`AssessmentRepository` `HistoryRepository`	Use Spring Data JPA for assessment and history queries.
`Criteria` · `AssessmentMapper`	Calculate scores and create safe response DTOs. Entities never leave the service.

`ReferenceDirectory` validates IDs through Feign; `FeignTokenRelay` carries the JWT. Names are saved as snapshots.

3. What is stored

Table in `HRMS.probation_schema`	Purpose
`probation_assessments`	IDs, dates, state, feedback, decision/publication, previous cycle and version.
`probation_assessment_ratings`	Criterion name/weight snapshots, 1–5 scores and evidence comments.
`probation_assessment_history`	Action, old/new status, actor, time, remarks and evidence snapshots.

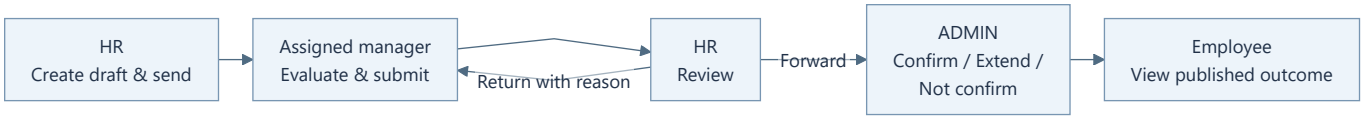
Flyway V1–V3 manage this schema; Hibernate validates it. Foreign keys are internal; external employee/user references are plain IDs.

4. Who can do what

`JwtAuthenticationFilter` verifies signature, expiry and identity claims. `CurrentUser` supplies the trusted identity; the backend enforces permissions.

Caller	Server-enforced access
HR	Creates/edits drafts, assigns managers, sends, reviews, returns or forwards; cannot decide.
Assigned manager	An enabled, employee-linked EMPLOYEE account. JWT user ID and employee link must match the stored assignment. No self-evaluation.
ADMIN	Views evidence/history, reassigns eligible reviews and decides after HR forwarding; does not author drafts.
Employee	Own period/status only; published outcome/comments after approval. No ratings or internal review notes.

From review evidence to a final outcome



HR can return an evaluation for correction. ADMIN's decision publishes the outcome immediately. An extension can have a separate linked follow-up cycle; the original stays closed.

5. Important APIs

Every path below starts with `/api/probation`; `{id}` is the assessment number.

Purpose	Method and path
HR setup	`POST /assessments` `POST /assessments/{id}/send-for-evaluation`
Manager	`PUT /assessments/{id}/evaluation` `POST /assessments/{id}/submit-evaluation`
HR review	`POST /assessments/{id}/return-to-manager` `POST /assessments/{id}/forward-for-approval`
Final decision	`POST /assessments/{id}/decision`
Read / queues	`GET /assessments`, `/assessments/{id}`, `/me`, `/manager/assignments` and `/statistics`

Lists support pagination and relevant filters. Errors use `{timestamp, status, error, message}`: 400 invalid input, 401 invalid identity, 403 denied access, 404 missing data, 409 conflict and 503 failed reference validation.

6. Example: Aditi's probation review

HR creates Aditi's 1 January–30 June assessment, sets a 15 June review date and assigns Kiran. He scores eight criteria and adds evidence, strengths, improvement areas, development advice and a recommendation. Eight scores of 4 produce **4.00/5** on the server. HR reviews and forwards it; ADMIN confirms Aditi and publishes approved comments. She sees the outcome, while internal discussion stays private. The score alone never confirms her.

7. Engineering decisions and validation

- **Separation:** probation owns its rules/data; Feign validates external IDs without querying another service's schema.
- **Assignment and states:** no global MANAGER role. `DRAFT` moves through evaluation, HR review and ADMIN approval. Wrong-state actions fail. Submission requires all scores, comments and feedback; the review date must be within the period.
- **Integrity and history:** versions prevent stale edits; PostgreSQL blocks active-period overlaps. Extension/non-confirmation requires reasons; extension needs a later end date. Final records are immutable. Each mutation is audited; reassignment archives earlier evidence.

8. Current scope and gaps

No runtime criteria editor, email/reminders or decision-amendment/cancellation workflow exists. Datasource credentials are inline, JWT has a demo-secret fallback, and issued tokens are not revoked immediately. These remain gaps; in-app queues are implemented.

9. A 60–90 second presentation script

> "Our probation service manages an employee's review through to a published decision. HR creates the assessment and assigns a reporting manager. Only that account can evaluate the employee. The manager scores eight criteria, adds evidence and recommends an outcome. The service calculates the score; it does not decide automatically.
> HR checks the submission and either returns it for correction or forwards it to ADMIN. ADMIN confirms, extends or does not confirm probation. Employees see their own status and published outcome, without confidential notes.
> React calls the gateway, which forwards requests to probation-service. The service verifies the JWT, checks permissions and workflow state, and saves changes in its PostgreSQL schema. Feign validates employee and reviewer references. Version checks protect concurrent edits, and audit history records who changed what. Final decisions stay closed; an extension can start a linked cycle. This module records the outcome without changing payroll or disabling accounts."